

Testing it: Fault Injection Service, Resilience Hub and ARC

Skill 2.1.3 – proving the plan works before the day it has to.

An untested runbook is not a plan. It is a **hypothesis** about what your systems will do.

Four ways to test, and they are not interchangeable

- ◆ **Tabletop** – talk through a scenario. Cheapest, finds gaps in roles, decisions and communication. Touches nothing.
- ◆ **Game day** – do it for real in a controlled window. Finds the renamed role and the deleted security group.
- ◆ **Purple team** – a red team attacks while the blue team responds, both knowing. Tests detection and response together.
- ◆ **Automated fault injection** – repeatable, scriptable, runnable in a pipeline. This is where FIS lives.
- ◆ Only the first is safe to run without preparation. The others can cause the incident they simulate.

ON THE EXAM

A tabletop finds out whether people know what to do. A game day finds out whether the systems still let them do it.

Fault Injection Service: the moving parts

- ◆ An **experiment template** holds actions, targets, stop conditions and an IAM experiment role.
- ◆ **Actions** are the faults – stop instances, throttle an API, add latency, pause EBS I/O, interrupt an AZ.
- ◆ **Targets** are selected by tag, resource ID or filter, so blast radius is chosen deliberately.
- ◆ **Stop conditions are CloudWatch alarms.** If one fires mid-experiment, FIS stops the experiment automatically.
- ◆ It can run **multi-account** experiments through target account configurations.

ON THE EXAM

Stop conditions are the feature that makes running this in production defensible. Learn them by name.

The scenario library, and its awkward edge

- ◆ Pre-built scenarios save designing experiments from scratch. **AZ Availability: Power Interruption** is the headline.
- ◆ It targets EC2, EBS, Auto Scaling groups, subnets, ElastiCache and RDS in one AZ.
- ◆ A **cross-Region connectivity** scenario tests one Region losing reach to another.
- ◆ **The scenario library is console-only.** Scenarios are not experiment templates and cannot be imported into FIS.
- ◆ For a pipeline, create a template from the scenario in the console, then export it.

ON THE EXAM

A console-only library in a service built for automation is real friction.

AWS Resilience Hub

- ◆ Assesses an application against a **resiliency policy** – your RTO and RPO – giving a score and component-level gaps.
- ◆ It **generates FIS experiment templates** from its recommendations, so assessment leads into a test.
- ◆ **Next generation GA May 2026** – app modelling, dependency discovery, GenAI failure assessment.
- ◆ **August 2026: recommended resilience tests** – pre-configured FIS tests, pass or fail.
- ◆ **It assesses and never remediates**, and the **score is point-in-time**.

ON THE EXAM

The point-in-time score is the trap. An old assessment describes an architecture that no longer exists.

Amazon Application Recovery Controller

- ◆ **Zonal shift** – you move traffic away from an impaired Availability Zone, on demand.
- ◆ **Zonal autoshift** – AWS does it when it detects impairment, and runs regular practice runs.
- ◆ **Readiness checks** continuously verify the standby can take traffic – capacity, quotas, configuration.
- ◆ **Routing controls and safety rules** – deliberate failover switches, guarded against failing into a worse state.
- ◆ FIS added a **recovery action** for zonal autoshift in March 2025.

ON THE EXAM

Autoshift's practice runs mean failover is exercised on a schedule, not first during an outage.

Resilience Hub, or Fault Injection Service?

RESILIENCE HUB

Should it survive?

- Assesses against stated RTO and RPO
- Produces a score and named gaps
- Generates FIS templates to test them
- Point-in-time; re-run after changes

FAULT INJECTION SERVICE

Does it survive?

- Injects real faults into real resources
- Stop conditions as CloudWatch alarms
- Repeatable, scriptable, multi-account
- Proves behaviour rather than predicting it

THE TIEBREAKER

Resilience Hub is theory. FIS is evidence. AWS now positions FIS as part of Resilience Hub, so expect them together.

But none of this tests your security response

- ◆ FIS injects infrastructure faults. It does not simulate an attacker.
- ◆ To test a detection and response pipeline, use **GuardDuty sample findings** – they flow through EventBridge like real ones.
- ◆ A sample finding proves the whole path works: rule, automation, OpsItem, page.
- ◆ For adversary behaviour, you need a **purple team exercise** or an authorised penetration test.
- ◆ AWS has a penetration testing policy. Some tests need prior authorisation.

ON THE EXAM

Know what the named services cover, and know what they do not.

A retailer has built an automated containment pipeline: GuardDuty finding, EventBridge rule, Step Functions workflow that snapshots volumes and moves the instance to a forensic security group, and an OpsCenter OpsItem for the on-call engineer.

It has never fired, because there has been no incident. The security manager wants proof the whole path works end to end, in production, without waiting for a genuine compromise and without any real resource being harmed.

How should the manager validate the pipeline?

- ☐ A Run an FIS experiment using the AZ Availability: Power Interruption scenario
- ☐ B Commission an authorised third-party penetration test against the production environment
- ☐ C Manually create a finding in Security Hub with the same finding type and severity
- ☐ D Generate GuardDuty sample findings and confirm each stage of the pipeline runs

◆ PAUSE AND DECIDE

How should the manager validate the pipeline?

- A

FIS AZ Power Interruption scenario

FIS injects infrastructure faults and generates no GuardDuty findings. It would test availability behaviour and never touch this pipeline.
- B

Third-party penetration test

It might eventually produce a finding, at high cost and slowly, with real risk to production. Enormous effort to test a rule you can trigger for free.
- C

Manually create a Security Hub finding

The near miss. It starts halfway along the path, so it never proves GuardDuty raised the finding or that the EventBridge pattern matches a real one.
- D

GuardDuty sample findings

Correct. Sample findings enter the pipeline at the true starting point and flow through EventBridge exactly like real ones, harming nothing.

THE SIGNAL

'End to end' rules out anything that starts partway along. Sample findings begin where a real finding begins.

A payments company wants to run FIS experiments in production during business hours to validate that its multi-AZ architecture and its incident runbooks behave as documented. The board has approved this only on condition that a degrading experiment cannot be allowed to become a real outage.

The platform team already has CloudWatch alarms on checkout success rate, p99 latency and payment API error rate.

What is the PRIMARY safety mechanism the team should configure?

- ☐ A Scope the FIS experiment role so it can only act on non-production resources
- ☐ B Define the existing CloudWatch alarms as stop conditions on the experiment template
- ☐ C Select targets by tag so that only a small subset of instances is affected
- ☐ D Schedule experiments through EventBridge so they run only in short windows

◆ PAUSE AND DECIDE

What is the PRIMARY safety mechanism the team should configure?

- A

 Scope the experiment role to non-production
Directly contradicts the requirement. The whole point is to run in production; restricting the role to non-production cancels the exercise.
- B

 CloudWatch alarms as stop conditions
Correct. A stop condition halts the experiment automatically when an alarm fires, which is exactly 'a degrading experiment cannot become a real outage'.
- C

 Limit targets by tag
Genuinely good practice and it limits the blast radius up front. It does nothing once the experiment is running and things start to degrade.
- D

 Schedule short windows via EventBridge
Bounds duration, not damage. A short window still lets an experiment run to completion while checkout is failing.

THE SIGNAL

'Cannot be allowed to become a real outage' describes something that reacts during the experiment. Only one option does.

A healthcare provider ran a Resilience Hub assessment fourteen months ago against a two-hour RTO and a fifteen-minute RPO. The application scored well and the report was filed as evidence for the annual audit.

Since then the team has migrated the database to a new engine, moved two components to a different Region, and replaced the load balancer. The compliance lead intends to submit the same report again this year, arguing the score is unchanged because no gaps were found.

What is the BEST response?

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| ☐ A Accept it. A resilience score remains valid until Resilience Hub reports new gaps | ☐ B Accept it, but run FIS experiments to confirm the score still holds |
| ☐ C Reject it. Scores are point-in-time and must be reassessed after architectural change | ☐ D Reject it, because Resilience Hub automatically remediates gaps and the report is now stale |

◆ PAUSE AND DECIDE

What is the BEST response?

- A** The score stays valid until new gaps are reported
Assumes Resilience Hub watches continuously. It does not. Nothing reassesses until somebody starts an assessment, so no new gaps will ever be reported.
- B** Accept it, but run FIS to confirm
Half right, and the half that is wrong is the order. Testing against objectives derived from an architecture that no longer exists validates the wrong thing.
- C** Reject it – scores are point-in-time
Correct. An assessment describes the architecture as it was that day. A new engine, a Region move and a new load balancer all invalidate it.
- D** Reject it – Resilience Hub auto-remediates
Right verdict, invented mechanism. Resilience Hub assesses and never remediates. It produces scores, gaps and FIS templates; engineers act on them.

THE SIGNAL

Two limits do the work here: the score does not auto-update, and the service never remediates. Option D gets one of them backwards.

An insurer runs a critical application across three Availability Zones behind an Application Load Balancer. During a recent AZ impairment the application degraded for 25 minutes while engineers manually drained the affected zone.

The insurer wants traffic moved away automatically when AWS detects an impairment, and wants regular proof that the mechanism still works rather than discovering it during the next real event.

What should the insurer implement?

- A Enable ARC zonal autoshift, and run the FIS AZ Availability: Power Interruption scenario to validate it
- B Configure ARC routing controls with safety rules that engineers trigger during an impairment
- C Create a Resilience Hub assessment with a 25-minute RTO and act on its recommendations
- D Configure Route 53 health checks with failover routing to the two healthy Availability Zones

◆ PAUSE AND DECIDE

What should the insurer implement?

- A

Zonal autoshift, validated with the FIS AZ Power Interruption scenario

Correct. Autoshift moves traffic when AWS detects impairment and runs practice runs; the FIS recovery action, added March 2025, proves the shift happens.
- B

ARC routing controls triggered by engineers

Still manual, which is the 25 minutes the insurer is trying to remove. Routing controls suit deliberate failover decisions, not automatic response.
- C

Resilience Hub assessment with a 25-minute RTO

Assessment tells them whether they are likely to meet an objective. It shifts no traffic, and 25 minutes is the failure they want to eliminate, not a target.
- D

Route 53 health checks with failover routing

Works across Regions or endpoints, not across AZs behind one ALB. The load balancer already owns AZ-level routing here.

THE SIGNAL

Two requirements – automatic shift and regular proof – and autoshift's practice runs plus the FIS recovery action answer both.

Five things to carry forward

- ◆ **Stop conditions are CloudWatch alarms.** The only FIS control that acts **during** an experiment.
- ◆ **Resilience Hub assesses; FIS proves.** Theory and evidence, and the guide names both.
- ◆ **Resilience Hub scores are point-in-time and it never remediates.** Reassess after any architectural change.
- ◆ **Zonal autoshift moves traffic automatically and practises on a schedule.** Routing controls are the manual equivalent.
- ◆ **FIS does not simulate attackers.** To test detection and response, use GuardDuty sample findings.

ON THE EXAM

Next: the incident itself. Containment and forensics – isolate, capture, eradicate.